

TWO-WHITES-الأبيضين

Beyond Whites · Sugar & Salt

Digital Self-Management Platform for Diabetes and Hypertension

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POLICY QUESTION

How might we provide continuous, systematic support for adults with diabetes and hypertension in Qatar to improve daily medication adherence and self-monitoring beyond clinical visits?

PROBLEM

Adults in Qatar with diabetes and hypertension lack continuous support to manage their conditions between clinical visits.

EVIDENCE

Adherence Gap

74%

Patients are non-adherent or delay medication

Health Impact

19%

of NCD deaths linked to diabetes & hypertension

Population Risk

24.3%

Hypertension prevalence in Qatar

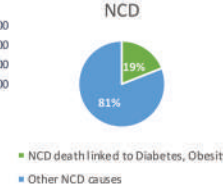
Key Context

- Non-adherence to medication or forget or delay medication — Qatar is high NCD burden in Qatar is high
- 74% of patients are non-adherent or delay medication
- 19% of NCD deaths are linked to diabetes & hypertension
- 24.3% hypertension prevalence in Qatar

NCD Mortality Comparison (Death per 100,000 of population)



Diabetes Contribution to NCD



PERSONA

Meet Ahmed, for whom we designed our policy specifically

Ahmed (41)

Ahmed (41)
Government employee, Doha
Type 2 diabetes & hypertension
Uses smartphone daily

"I know I should take it every day — I just keep forgetting."



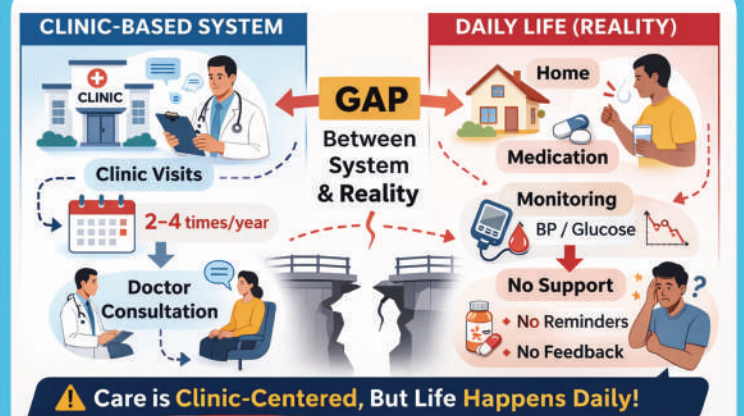
A Typical Day (Without Support System)

7:00 AM	No reminder system in place — starts the day off-track.
8:30 AM	Busy commute, no quick monitoring tool available
2:00 PM	Takes medication late (Inconsistent timing reduces treatment effectiveness)
8:00 PM	Fatigue and no evening reminder led to another missed dose
3 months later:	No monitoring record means the doctor cannot track Ahmed's progress

Key Pain Points

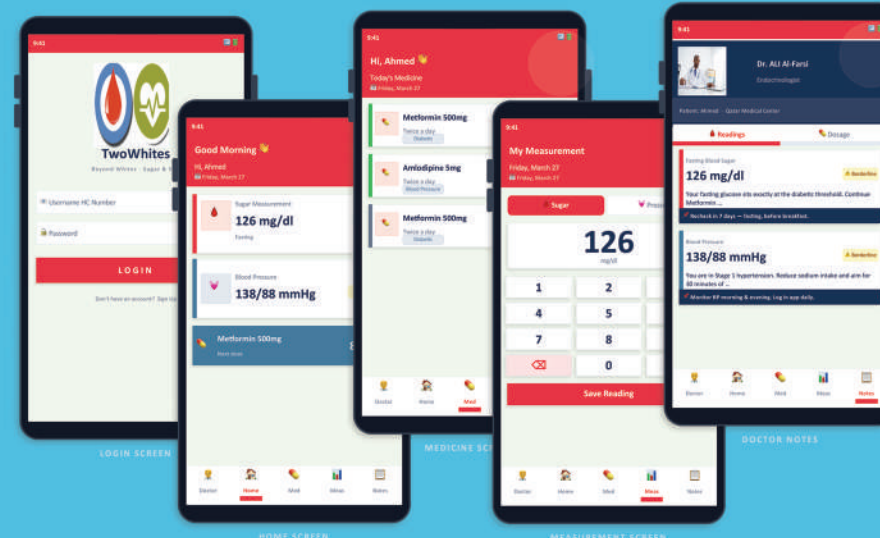
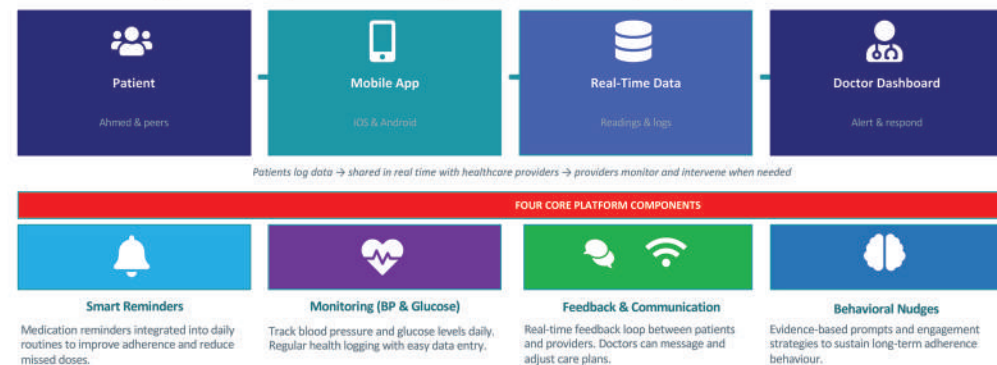
- No reminders integrated into daily routine
- No real-time feedback on health status
- Monitoring is effortful and inconsistent
- Doctor cannot track progress between visits

SYSTEM GAP



SOLUTION — DIGITAL SELF-MANAGEMENT PLATFORM

A digital platform enabling continuous DM & HTN self-management beyond clinical visits — connecting patients to providers every day, not just 4x per year.



IMPLEMENTATION PLAN — OPTION B (MYHEALTH INTEGRATION)

3-4 months	2-3 months	3-6 months	Ongoing
Phase 1 — SYSTEM INTEGRATION Actors: E-Health · MPH · HMC · Vendors Integrate self-management module into MyHealth Ensure interoperability with PHCC & HMC EMR systems Connect PHCC and HMC via national platform Validate clinical indicators and system design	Phase 2 — WORKFORCE ENABLEMENT Actors: PHCC · HMC · E-Health Train PHCC & HMC staff on system use Integrate tool into routine care workflows Prepare clinical protocols & monitoring procedures Ensure system usability and technical support	Phase 3 — NATIONAL DEPLOYMENT Actors: PHCC · E-Health · MoPH Roll out system through MyHealth platform Enrol patients through PHCC systems nationally Activate reminders, tracking & monitoring features Enable real-time data sharing with providers	Phase 4 — MONITORING & EVALUATION Actors: HBKU · MPH · PHCC Monitor adherence & usage through dashboards Track real-time KPIs (adherence rate, active users) Generate quarterly reports for policy review Continuously improve the system based on evidence

Total estimated timeline: 8–13 months to full deployment · Then ongoing monitoring & evaluation

INPUTS

- MOPH funding, leadership & regulatory authority
- MyHealth platform infrastructure
- E-Health governance & interoperability frameworks
- PHCC patient access & enrolment systems
- HMC clinical data systems
- Contracted digital health vendors
- Evaluation & research capacity (HBKU)

ACTIVITIES (CLINICAL)

- HMC integrates patient data across care levels
- HMC validates clinical indicators and outcomes.
- PHCC & HMC connect systems through a national platform.
- PHCC and HMC train health-care providers on digital workflows
- PHCC enrolls patients through routine care.
- PHCC monitors medication adherence and provides follow-up care
- HBKU and HMC conduct evaluation and research

ACTIVITIES (DIGITAL)

- E-Health integrates self-management module into MyHealth or creates standalone app
- E-Health ensures usability & technical support
- E-Health tests, pilots & refines systems
- E-Health enables real-time monitoring & reporting
- MOPH ensures EMR interoperability across
- PHCC & HMC
- MPH establishes national digital self-management policy

OUTPUTS

(System): Self-management digital platform (integrated within MyHealth or deployed as a standalone application) successfully developed, tested, and piloted across 3 national primary care facilities within 18 months of launch

(Engagement): "40–60% of enrolled patients with Diabetes and Hypertension engaged the platform at 12 months, (medication reminders, self-monitoring entry, or clinical messaging)

(service delivery): Real-time monitoring and reporting system operational across PHCC and HMC, with EMR interoperability established and clinical staff trained on digital workflow integration.

OUTCOMES

Short-term: 65–75% for medication adherence; 55–65% for consistent self-monitoring, reported separately between 1–12 months

Long-term: 10–20% reduction in acute complication events (hospitalisations, emergency presentations), microvascular complication reduction of 20–30% at 10+ years,

IMPACT

20–30% reduction in NCD-related mortality" with "contribution to Qatar's national NCD strategy targets,

BUDGET

OPTION A — Standalone App		OPTION B — MyHealth PHCC Integration		
QAR 620,000	18–24 months	QAR 330,000	12–18 months	
TOTAL BUDGET	TIMELINE	TOTAL BUDGET	TIMELINE	
✔ Option B saves QAR 290,000 vs Option A — Fully justified by existing MyHealth infrastructure, PHCC-grade security reuse & shared maintenance				
Phase Alignment Key:		Phase 4 — Monitoring & Eval.		
Phase 1 — Setup / Integration		Phase 3 — Deployment		
Phase 2 — Pilot / Workforce		Phase 1 — Phase		
Cost Category	Option A — Standalone	Option B — MyHealth	Savings	Phase
 App / Module Development iOS/Android app vs MyHealth module	QAR 350,000 	QAR 150,000 	Save QAR 200K	Ph. 1
 Doctor Dashboard Web portal vs Integrated dashboard	QAR 80,000 	QAR 40,000 	Save QAR 40K	Ph. 1
 API Integration PHCC API vs MyHealth CDS/EMR (existing)	QAR 60,000 	QAR 50,000 	Existing API	Ph. 1
 Data Security & Compliance Full compliance vs PHCC-grade (reused)	QAR 50,000 	QAR 20,000 	Save QAR 30K	Ph. 1
 System Maintenance (Year 1) Dedicated staff vs Shared infrastructure	QAR 50,000 	QAR 40,000 	Shared Infra	Ph. 4
 Training & Onboarding New hires + staff training — Both options	QAR 30,000 	QAR 30,000 	Same	Ph. 2-3
TOTAL	QAR 620,000 18–24 months	QAR 330,000 12–18 months	—QAR 290K	

STAKEHOLDERS AND POLICY FRAMEWORKS

